INTRODUCTION TO DATA MINING

Table of contents :

Data :

- In data mining, data refers to the information (or) facts stored in a database (or) any source
- It can be numbers, text, images (or) any kind of information collected from the real world

Ex :

- Imagine a supermarket keeps records of what customers buy everyday
- These records (like "customer A bought apples and milk") are the data
- Data mining uses this data to find patterns like " Most customers buy milk with bread "
- This helps the supermarket make better decisions such as putting milk and bread close to each other

Types of data :

- In data mining, data can come in different types based on how it is organized, structured, or represented
- Here are some common types of data are
    - Structured data
    - Unstructured data
    - Semi-structured data
    - Nominal (or) categorical data
    - Ordinal data
    - Interval data
    - Ratio data
    - Time series data
    - Spatial data

Structured Data :

- This type of data is highly organized and easy to analyze
- It usually fits into rows and columns, like a table or spreadsheet

Ex :

- Customer names, purchase amounts, dates of purchases, etc. (Think of a database or Excel sheet.)

Unstructured Data :

- This data doesn't have a predefined structure, so it's harder to analyze directly.

Ex :

- Text files, social media posts, videos, images and so on

Semi-Structured Data :

- This data has some structure but doesn't fit perfectly into a traditional database like structured data
- It may have tags (or) markers to separate elements

Ex :

- XML files, JSON files where the data is organized but not as strictly as in a table

Nominal Data (Categorical Data) :

- This is data that represents categories (or) groups without any order

Ex :

- Gender, color, country and type of product (like "Shirt", "Pants", "Shoes").

Ordinal Data :

- This data has a clear order, but the difference between values isn’t necessarily uniform or measurable.

Ex :

- Customer satisfaction ratings (e.g., "Very Satisfied", "Neutral", "Dissatisfied").

Interval Data :

- This data has both order and consistent differences between values, but there is no true zero point

Ex :

- Temperature in Celsius (or) Fahrenheit (where the difference between values is meaningful, but zero doesn’t represent the absence of temperature).

Ratio Data :

- This data has both order and consistent differences, and it has an absolute zero point, meaning zero represents the absence of the value.

Ex :

- Age, income, or sales amounts (where zero truly means "none" or "nothing").

Time-Series Data :

- This is data that is collected over time, often at regular intervals

Ex :

- Stock prices, weather data, or monthly sales numbers.

Spatial Data :

- This data represents information about physical locations and their properties.

Ex :

- GPS coordinates location data from smart phones (or) maps

Mining :

- Mining means digging through large amounts of data to find useful patterns, trends (or) information just like digging for gold in a mine
- It's about extracting valuable knowledge from the raw data

Ex :

- Imagine you have a school database with student's exam scores
- Mining would involve analyzing this data to find patterns like " student who study for more than 2 hours daily score above 80% "
- This process helps you discover hidden insights from the data

What is a data mining :

- Data mining is the process of extracting knowledge from large amounts of data using various statistical and computational techniques
- The data can be structured, semi-structured and unstructured
- The data can be stored in various forms such as databases, data warehouses and data lakes
- The primary goal of data mining is to discover hidden patterns (or) useful information from large sets of data
- It's like finding hidden gems in a huge pile of data

Ex :

- Let's say a store sells umbrellas
- By looking at their sales data, they notice that umbrellas sell a lot more on rainy days than on sunny days
- This pattern is something the store can discover using data mining

- Once they know this, they can prepare better by stocking more umbrellas before it rains, helping them sell more
- Data mining helps the store find out that people buy more umbrellas when it rains

Applications of data mining :

- There are some important applications of data mining
- Some of the applications are
    - Market basket analysis
    - Customer segmentation
    - Recommendation systems
    - Financial marketing forecasting
    - Healthcare
    - Churn prediction
    - Credit scoring
    - Agriculture

Market basket analysis :

- Retailers use data mining to identify the products frequently purchased in combination
- This supports targeted marketing, product placement and store design

Customer segmentation :

- Organizations use data mining to group customers who have similar behaviours
- For example, they might notice that some customers like to buy certain products (or) they shop at specific times
- By finding these patterns, the company can create special offers (or) product recommendations just for those groups
- So, if a store know that a group of customers loves buying shoes every month, they might send them special discounts on shoes (or) recommend new styles
- This helps the store market in a more personalized way, making customers feel like the offers are just for them

Recommendation systems :

- Online stores like Amazon (or) Netflix suggest products (or) movies based on past behaviours and preferences

Financial marketing forecasting :

- In finance, data mining helps predict things like stock prices, currency exchange rates, market trends by looking at past data

Healthcare :

- Hospitals and researchers can use data mining to identify patterns in patient data, helping predict diseases (or) conditions before they develop
- Data mining is used to analyze medical images (like X-rays (or) MRIs) to detect signs of diseases like cancer

Churn prediction :

- Telecom companies use data mining predict which customers are likely to cancel their services, allowing them to take action to retain them

Note :

- Churn refers to the number of customers (or) users who stop using a product (or) service over a certain period of time

Credit scoring :

- Financial institutions like banks, use data mining to decide if they should lend money to someone (or) a company
- They look at things like a person's past payments, income, debts and other financial details to figure out how risks it is to lend money to them

Agriculture :

- Farmers use data mining to help them grow more crops and use fewer resources like water and fertilizer
- They do this by looking at data on things like how crops have grown in the past, weather and the condition of the soil

Advantages of data mining :

- Data mining offers several advantages, making it a valuable tool for businesses and organizations
    - Finds hidden patterns
    - Helps make better decisions
    - Improved product development
    - Saves time and money
    - Improves customer experience
    - Predicts future trends

Finds hidden patterns :

- Data mining can uncover hidden patterns that you might not see right away
- For example, a store might find that people who buy milk often also buy bread, helping them organize products better

Helps make better decisions :

- By analyzing data, businesses can make smarter decisions
- For example, a company might use data mining to figure out which product are most popular, so they can stock up on those items

Improved product development :

- Analyzing customer feedback, purchase history and market trends allows businesses to design products (or) services that better meet the needs of their customers, leading to more successful offerings

Saves time and money :

- Data mining helps organizations quickly analyze large amounts of data, saving time compared to doing it by hand
- For example, a bank might use data mining to spot fraudulent transactions without having to check one manually

Improves customer experience :

- Businesses can use data mining to understand customers needs and personalize their services
- For instance, a streaming service might recommend shows based on what you have watched before

Predicts future trends :

- Data mining can help predict what might happen in the future
- For example, a clothing store could use past sales data to predict what types of clothes will be popular next season

Disadvantages of data mining :

- The disadvantages of data mining are
    - Privacy concerns
    - Data misinterpretation
    - Costly to implement
    - Ethical issues

Privacy concerns :

- Data mining often involves analyzing personal information
- This can lead to privacy issues if sensitive data is used without permission
- For example, if a company collects data about your shopping habits without your consent, it could feel invasive

Data misinterpretation :

- Sometimes, the patterns found through data mining might be misleading (or) incorrect
- For example, a store might think that a drop in sales is because of a certain product, but it could actually be due to something else like a holiday

Costly to implement :

- Setting up data mining systems can be expensive, especially for small businesses
- They need special software, skilled staff and time to analyze the data, which can be a big investment

Ethical issues :

- Sometimes, businesses use data mining in ways that are not ethical like targeting vulnerable customers with too many ads (or) manipulating prices based on someone's buying habits

Data mining challenges :

- Even though data mining is a powerful technique, it faces several challenges are
    - Incomplete and noisy data
    - Data distribution
    - Complex data
    - Performance issues
    - Data privacy and security
    - Data visualization

Incomplete and Noisy Data :

- Real-world data is often messy (not clean), missing (or) incorrect
- Mistakes happen due to human error (or) system issues
- For example, a store collects customer phone numbers for purchases over $500. If an employee types a wrong digit, the data becomes incorrect and some customers might also refuse to share their number, making the data incomplete

Data Distribution :

- Data is often stored in different places (like separate servers or the internet)
- Combining all this data into one system is difficult due to technical and privacy concerns

- For example, a company has multiple offices, each storing data on its own server. It’s not practical to move all data to one central server, so special tools are needed to analyze data from different locations

Complex Data :

- Data comes in different formats like text, images, videos and audio
- Extracting useful information from all these different types is challenging
- For example, a social media platform stores text posts, photos and videos and analyzing this mixed data to find trends (like which posts are most popular) requires advanced tools

Performance Issues :

- The speed and accuracy of data mining depend on how well the algorithms are designed
- Poorly designed methods make data mining slow and inefficient
- For example, If a bank uses a slow algorithm to detect fraud, it might take too long to identify suspicious transactions, allowing fraudsters to withdraw money before being caught

Data Privacy and Security :

- Data mining can reveal personal information without users' consent, leading to privacy concerns
- For example, If a grocery store analyzes purchase records, they might learn a customer's shopping habits without their permission, which could feel like an invasion of privacy

Data Visualization :

- After mining data, the results must be shown in a clear and understandable way
- However, presenting complex data in a simple, user-friendly format is difficult
- For example, a sales report that shows trends with complicated numbers and graphs may confuse business managers. If displayed as an easy-to-read chart, it would be more useful

Why data mining :

- We live in a world where huge amounts of data are collected everyday from online shopping, social media, bank transactions, healthcare and so on
- Analyzing this data is very important to make better decisions and improve services

Mining toward the information age :

- “ We are living in the information age ” is a popular saying but in reality, we live in the data age because we generate and store huge amounts of data every day
- Stores like Wal-Mart handle millions of sales transactions weekly
- Satellites, experiments and machines generates huge data sets continuously
- Hospitals store patient records, test results and medical images daily
- People upload millions of pictures, videos and posts everyday
- Billions of Google searches and phone calls create massive data traffic

- We have huge amounts of data, but without proper tools, it is useless
- Businesses organizations struggle to make sense of all this data
- Now data mining is like a smart detective – It analyzes large data sets to find patterns and useful information automatically

<u>Data mining as the evolution of information technology :</u>

- Data mining is a natural step in the growth of information
- As technology improved businesses collected more and more data
- But having a lot of data was not enough, they need a way to understand and use that data effectively
- This led to the development of data mining

<u>How did data mining evolve :</u>

<u>Early days (1960's – 1980's) :</u>

- Data was stored using simple file systems
- Databases evolved from hierarchical to relational database (tables)
- People could store and retrieve data, but analyzing it was difficult

<u>Database systems (1980's – 1990's) :</u>

- Relational databases became popular [i.e., MySQL, Oracle]
- Query languages (SQL) allowed users to access data more easily
- Online transaction processing (OLTP) improved data management

<u>Advanced data analysis (1990's - Present) :</u>

- Data warehouses were introduced to store and manage large datasets
- OLAP (Online Analytical Processing) tools helped analyze data in multiple ways
- Data mining emerged to find patterns, trends and useful insights

<u>What is data mining :</u>

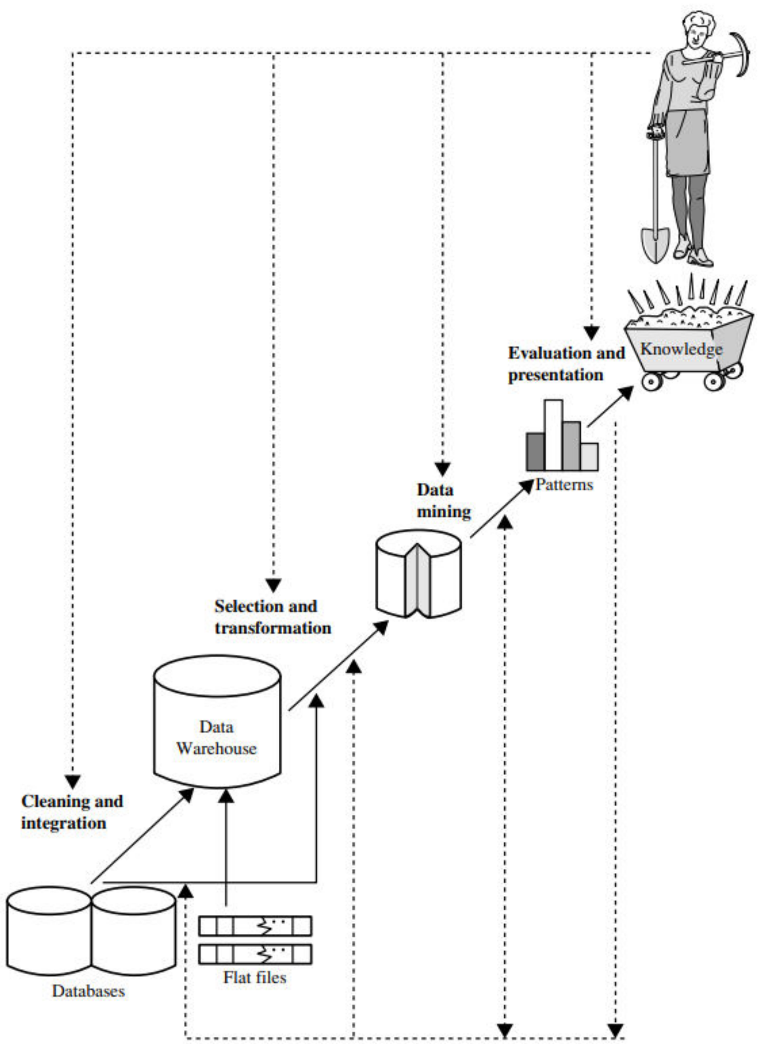
- Data mining is the process of extracting knowledge from large amounts of data using various statistical and computational techniques
- The data can be structured, semi-structured and unstructured
- The data can be stored in various forms such as databases, data warehouses and data lakes
- The primary goal of data mining is to discover hidden patterns (or) useful information from large sets of data
- It's like finding hidden gems in a huge pile of data

Ex :

- Let's say a store sells umbrellas
- By looking at their sales data, they notice that umbrellas sell a lot more on rainy days than on sunny days
- This pattern is something the store can discover using data mining
- Once they know this, they can prepare better by stocking more umbrellas before it rains, helping them sell more
- Data mining helps the store find out that people buy more umbrellas when it rains

Knowledge discovery in databases :

- KDD means Knowledge Discovery in Databases
- KDD is the process of finding useful information from a large amount of data
- It helps turn raw data into the meaningful data



Data cleaning :

- Data cleaning is the first step of KDD process in data mining
- Data cleaning is defined as removal of noisy and irrelevant data from collection

(or)

- Data cleaning is the process of fixing (or) removing incorrect, incomplete, duplicate and inconsistent data to improve its quality
- It helps to make the data accurate and useful for analysis
- Data cleaning using data cleaning tools like excel, talend and so on

Ex :

- Before data cleaning

Name	Age	Email	Purchase Amount ($)
John	25	john@email.com	50
Sarah	?	sarah@email.com	100
Mike	30	-	200
Anna	28	anna@email.com	0
John	25	john@email.com	50

- After data cleaning

Name	Age	Email	Purchase Amount ($)
John	25	john@email.com	50
Sarah	27	sarah@email.com	100
Mike	30	mike@email.com	200
Anna	28	anna@email.com	50

Note :

- Noisy data means data that has errors, irrelevant information (or) random variations that make it unclear (or) inaccurate
- It can be caused by mistakes in data entry, sensor errors and so on

Data integration :

- Data integration is defined as heterogeneous data from multiple sources combined in a common source [i.e., Data Warehouse]
- Data integration using data migration tools like ETL (Extract-Load-Transformation) process

Ex :

- Imagine a company has customer data stored in different places
    - Sales database [i.e., Contains customer names and purchases]

- Support system [i.e., Contains customer complaints]
- Marketing list [i.e., Contains customer E-Mail and preferences]

- If these data sources are integrated, the company can get a complete view of each customer – knowing what they bought, what issues they had and what promotions they are interested in
- This helps in better customer service and decision making

Data selection :

- Data selection is defined as the process where data relevant to the analysis is decided and It is retrieved from the data collection
- It helps remove unnecessary data and focus only on useful information
- Data selection using neural networks, decision trees, naïve bayes, clustering, regression and so on

Ex :

- Suppose a company has a customer database with the following details

Name	Age	Email	City	Last Purchase ($)	Subscription Status
John	25	john@email.com	New York	100	Active
Sarah	30	sarah@email.com	Los Angeles	50	Inactive
Mike	40	mike@email.com	Chicago	200	Active

- If the company only wants data of active subscribers
- They will select only the rows where subscription status = active and ignore others
- After data selection, it shows likes this

Name	Age	Email	City	Last Purchase ($)	Subscription Status
John	25	john@email.com	New York	100	Active
Mike	40	mike@email.com	Chicago	200	Active

- This process helps in efficient analysis by focusing only on necessary data

Data transformation :

- Data transformation is defined as the process of transforming data into appropriate form required by mining procedure

Ex :

- Suppose a company collects customer data in different formats

Name	Age	Purchase Amount ($)	Date of Purchase
John	25	1,000.50	01-02-24
Sarah	30	500.75	05-02-24
Mike	40	$2,000	10-Feb-24

- In the above table you have observed that inconsistent date formats and different currency formats
- After data transformation, it shows likes this

Name	Age	Purchase Amount ($)	Date of Purchase
John	25	1000.5	01-02-24
Sarah	30	500.75	05-02-24
Mike	40	2000	10-02-24

Data mining :

- Data mining is the process of finding useful patterns, trends and insights from large amounts of data
- It is a key step in KDD (Knowledge discovery in databases), which is the overall process of turning raw data into meaningful information

Ex :

- Suppose a supermarket only has basic transaction records without any insights (Without depth knowledge)

Customer	Items Purchased	Total Spend ($)
John	Bread, Milk, Butter	15
Sarah	Bread, Milk	10
Mike	Bread, Butter	12
Anna	Milk, Butter	14

- After data mining, it shows likes this

Customer	Items Purchased	Total Spend ($)	Discount Applied
John	Bread, Milk, Butter	15	-$1 on Butter
Sarah	Bread, Milk, Butter (Suggested)	13 (More Sales)	-$1 on Butter
Mike	Bread, Butter	12	-$1 on Butter
Anna	Milk, Butter	14	-$1 on Butter

Pattern evaluation :

- Pattern evaluation is the process of checking which patterns found in data mining are actually useful and meaningful

- It helps remove unimportant (or) misleading patterns and keeps only the most valuable ones

Ex :

- Suppose a shopping website uses data mining to find buying patterns and it discovers these
  - " Customers who buy laptops often buy laptop bags " [i.e., Useful]
  - " Customers who buy shoes also buy mobile chargers " [i.e., Not useful, random pattern]
  - " Customers who buy baby products also buy diapers " [i.e., Useful]
- The useful patterns 1 and 3 are kept for marketing strategies
- The random pattern 2 is removed because it doesn't make sense

Knowledge representation :

- Knowledge representation is the final step in the KDD process
- After finding useful patterns from data, we need to present them clearly using tables, graphs, charts and so on in knowledge representation

Ex :

- " Customers who buy mobile phones often buy earphones and phone cases "
  - Table – A list showing how many customers buy these items together
  - Graph / Chart – A bar chart showing the percentage of mobile buyers who also buy earphones

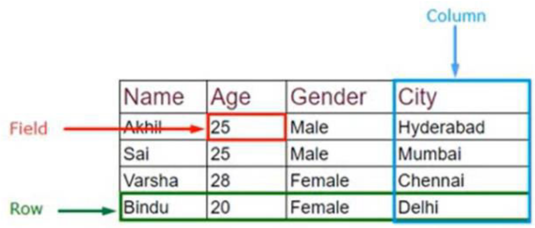
What kinds of data can be mined :

- Data mining is a technology that can be used on any kind of data, as long as the data is useful for a particular task
- The most common types of data used for data mining. They are
  - Database data (or) Relational database
  - Data warehouse data
  - Transactional data
  - Other kinds of data

Database data :

- A database system (or) DBMS is a software that helps store, manage and retrieve data efficiently
- It acts like a digital filing cabinet where data is stored in an organized way
- Relational database is a collection of tables, each of which is assigned a unique name, each table consists of a set of attributes (columns (or) fields) and usually stores large set of tuples (records (or) rows)
- Each tuple in a relational table represents an object identified by a unique key and described by a set of attribute values

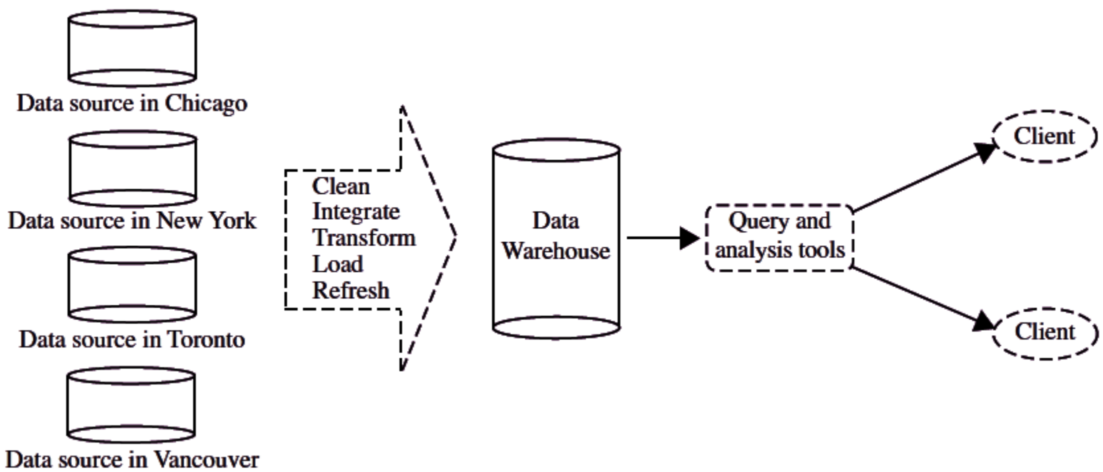
Ex :



Name	Age	Gender	City
~~Akhil~~	25	Male	Hyderabad
Sai	25	Male	Mumbai
Varsha	28	Female	Chennai
Bindu	20	Female	Delhi

Data warehouses data :

- Data warehouse is a repository of information collected from multiple sources, stored under a unified schema and usually residing at a single site
- Data warehouses are constructed via a process of data cleaning, data integration, data transformation, data loading and periodic data refreshing



Data sources :

- Data is collected from different locations like Chicago, New York, Toronto and Vancouver
- These could be different branch offices of a company, sales data from stores, online transaction records and so on

Data pre-processing :

- Before storing in the data warehouse, the data goes through several steps
    - Cleaning – Removing errors (or) inconsistencies
    - Integration – Combining data from different sources into a unified format
    - Transformation – Converting data into a structured format
    - Loading – Storing the processed data in the warehouse

- Refreshing – Updating the warehouse with new data regularly

Data warehouse :

- A central storage where all the cleaned and structured data is kept
- This acts as a single source of truth for businesses

Query and analysis tools :

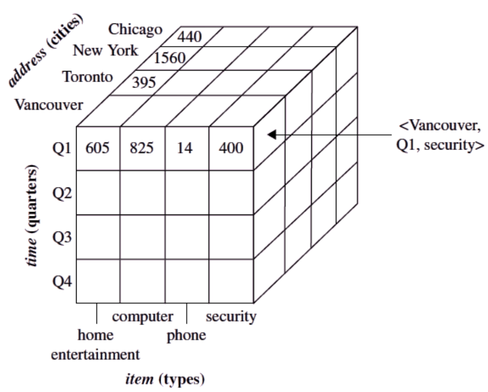
- Special tools help in analyzing the data and generating reports

Clients :

- These are business users (or) decision-makers who use the processed data for insights
- For example, a manager can check which city has the highest sales and plan marketing strategies accordingly

Ex :

- A data warehouse is usually modelled by a multidimensional data structure called a data cube, in which each dimension corresponds to an attribute (or) set attributes in the schema and each cell stores the value of some aggregate measure such as count (or) sum
- A data cube provides a multidimensional view of data and allows the pre-computation and fast access of summarized data



Transactional data :

- Transactional database is a collection of data organized by time stamps, date and so on to represent transaction in databases
- In general, each record in a transactional database captures a transaction such as customer's purchase, flight booking and so on
- A transaction typically includes a unique transaction identity number (Trans ID) and a list of the items making up the transaction, such as the items purchased in the transaction

- This type of database has the capability to roll back (or) undo operation when a transaction is not completed (or) committed and it follows ACID property of DBMS

Ex :

TID	Items
T1	Bread, Coke, Milk
T2	Popcorn, Bread
T3	Popcorn, Coke, Egg, Milk
T4	Popcorn, Bread, Egg, Milk
T5	Coke, Egg, Milk

Other kinds of data :

Multimedia database :

- Multimedia databases are used to store multimedia data such as images, animation, audio, video along with text
- This data is stored in the form of multiple file types like .txt(text), .jpg(images), .swf(videos), .mp3(audio) and so on

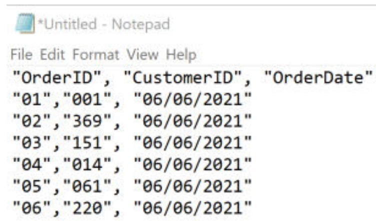
World wide web :

- World wide web is a collection of documents and resources such as audio, video, text
- It identifies all this by URLs of the web browsers which are linked through HTML pages
- It is the most heterogeneous repository as it collects data from multiple resources and it is dynamic in nature as volume of data is continuously increasing and changing

Text data (or) flat file :

- Flat files are a type of structured data that are stored in a plain text format
- They are called " flat " because they have no hierarchical structure, unlike a relational database table
- Flat files typically consists of rows and columns of data with each row representing a single record and each column representing a field (or) attribute within that record
- They can be stored in various formats such as Comma-Separated Values (.CSV), Tab-Separated Values (.TSV) and so on
- Flat files is defined as data files in text form (or) binary form with a structure that can be easily extracted by data mining algorithms
- Data stored in flat files have no relationship (or) path among themselves, like if a relational database is stored on flat file, then there will be no relations between the tables

Ex :



*Untitled - Notepad

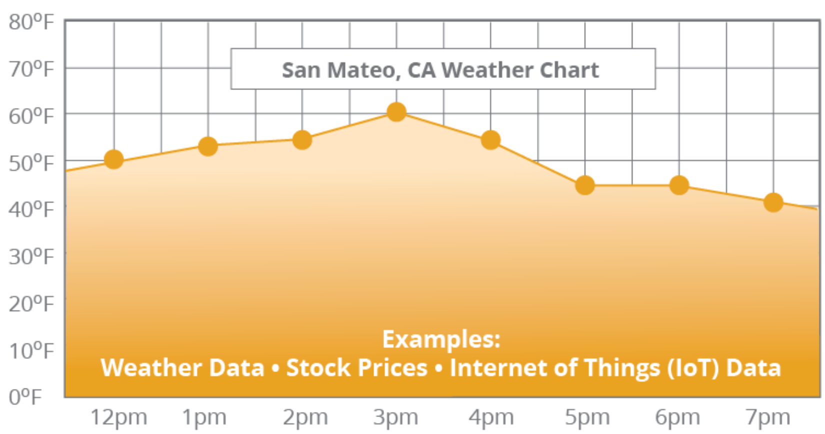
File Edit Format View Help

```
"OrderID", "CustomerID", "OrderDate"
"01","001", "06/06/2021"
"02","369", "06/06/2021"
"03","151", "06/06/2021"
"04","014", "06/06/2021"
"05","061", "06/06/2021"
"06","220", "06/06/2021"
```

Time series database :

- Time series data is a sequence of data points collected over time intervals, allowing us to track changes over time
- Time series data can track changes over milliseconds, days, years
- Time series data are simply measurements (or) events that are tracked, monitored and aggregated over time
- This could be application performance monitoring, network data, sensor data, trades in a market and many other types of analytics data

Ex :



A Time Series Database is a database that contains data for each point in time.

Data mining tasks :

- Data mining task is a specific goal (or) action that helps us find useful patterns, relationships (or) information from large amounts of data
- It helps businesses, scientists and other professionals make better decisions using data
- Data mining tasks are two types. They are
    - Descriptive tasks
    - Predictive tasks

Descriptive tasks :

- Descriptive tasks are helps us to understand what is happening in the data

Ex :

- A supermarket studies customer purchases and finds that people who buy bread often buy butter too
- This helps in placing products together to boost sales

Types of descriptive tasks :

- They are 4 types of descriptive tasks
    - Clustering
    - Association rules
    - Summarization
    - Sequence discovery

Clustering :

- Grouping similar data together
- For example, grouping customers based on shopping behaviour [i.e., Budget shoppers vs Luxury shoppers]

Association rules :

- Finding relationships between items
- For example, Amazon suggesting " customers who bought a phone also bought a phone case "

Summarization :

- Creating an overview of data
- For example, a university analyzes student performance data to summarize overall academic trends

Sequence discovery :

- It is used to find patterns in ordered events (or) actions overtime
- It helps to identify common sequences that frequently occur in data
- For example, a shopping website analyzes customer behaviour and finds this pattern – Customer first search for a mobile phone → Then check reviews → Then add to cart → Finally make a purchase
- This sequence helps the website improve recommendations, offer discounts at the right time and increase sales

Predictive tasks :

- Predictive tasks are helps us to understand to guess what will happen in the future
- For example, a bank checks a person's past loan repayment history and predicts whether they will repay a new loan

Types of predictive tasks :

- They are 4 types of predictive tasks
    - Classification
    - Prediction
    - Time series analysis
    - Regression

Classification :

- Categorizing data into predefined groups
- For example, an email service classifies emails as “ spam ” (or) “ Not spam ”

Prediction :

- Prediction means using past data to guess what will happen in the future
- It helps in decision making by estimating outcomes before they happen
- For example, a bank checks a customer’s past loan payments. If they have always paid on time, the bank predicts that they will repay a new loan as well

Time series analysis :

- Time series analysis is studying data collected overtime to find patterns, trends and make predictions
- It helps understand how things change over time
- For example, a weather station records the temperature every day. By analyzing past temperature data, they can identify trends [i.e., Summer gets hotter] and predicts future temperatures

Regression :

- Predicting numerical values
- For example, a real estate company predicts house prices based on location and size

Data mining functionalities :

- The following are the data mining functionalities
    - Characterization and discrimination
    - Mining frequent patterns, associations and correlations

- Classification and regression for predictive analysis
- Cluster analysis
- Outlier analysis

Characterization and discrimination :

- In data mining characterization and discrimination are two ways to understand and analyze data

Data characterization :

- Data characterization is a way of summarizing the key features of a specific group of data
- It helps us understand the general patterns and trends in the data
- We use graphs, charts like bar charts and pie-charts to describe the data

Ex :

- Imagine you own a toy store and you want to know what kind of customers spend the most money
- You analyze your sales data and find that
    - Most high-spending customers are between 30 and 40 years old
    - They usually buy toys for their children
    - They prefer educational toys
    - They often shop online rather than in the physical store
- This summary helps you understand your best customers, so you can make better business decisions like offering discounts on educational toys (or) improving your online store

Data discrimination :

- Data discrimination is a way of comparing two different groups of data
- It helps find differences between them by looking at their characteristics
- It uses rules to compare one group with another and highlight what makes them different

Ex :

- Imagine you own a coffee shop and you want to compare regular customers (who visit at least twice a week) with occasional customers (who visit only a few times a year)
- After analyzing the data, you find
    - Regular customers are mostly young professionals (25-35 years old), prefer black coffee and often use the café's free Wi-Fi to work
    - Occasional customers are mostly older adults (50+ years old), prefer plain coffee and usually come in the mornings
- This comparison helps you make better decisions like offering discounts on black coffee for young professionals (or) a special morning deal for older adults

Mining frequent patterns, associations and correlations :

- Frequent patterns are patterns that show up often in data
- Think of them as things that frequently happen together
- The frequent patterns are three types
    - Frequent item sets
    - Frequent sub sequences
    - Frequent sub structures

Frequent item sets :

- These are groups of things that often appear together
- For example, In a grocery store, people often buy milk and bread together. So, milk and bread is a frequent item set

Frequent sub sequences : (Sequential patterns)

- These are things that happen in a certain order, one after another
- For example, a customer first buys a laptop, then a digital camera and later a memory card. This buying pattern is a frequent sequence

Frequent sub structures :

- These are patterns in complex structures like networks, trees, graphs
- For example, In a social networks, a common friend connection pattern might be – A is friends with B, B is friends with C and C is also friends with A. If this structure appears often, it's a frequent substructure

Association analysis :

- Association analysis is used to find relationships between different items in a dataset
- It helps discover patterns like which products are often bought together
- It looks at past transaction to find rules like support and confidence
- Support means " How often a combination appears in all transactions "
- Confidence means " The probability that when one item is bought, another related item is also bought "

Ex :

- Suppose an example of such rule, mined from the All Electronics transactional database is

$$buys(X, \text{"}computer\text{"}) \Rightarrow buys(X, \text{"}software\text{"})\ [support = 1\%, confidence = 50\%]$$

- Where X is a variable representing a customer

- 1% support means that 1% of all transactions under analysis show that computer and software are purchased together
- 50% confidence (or) certainty means that customer buys a computer, there is a 50% chance that she will buy software as well
- This association rules involves a single attribute (or) predicate (i.e., buys) that repeats
- Association rules that a single predicate are referred to as a single dimensional association rules

Note :

- Multi-dimensional association rule

$$age(X, \text{"20..29"}) \wedge income(X, \text{"40K..49K"}) \Rightarrow buys(X, \text{"laptop"})$$
$$[support = 2\%, confidence = 60\%].$$

Classification and regression for predictive analysis :

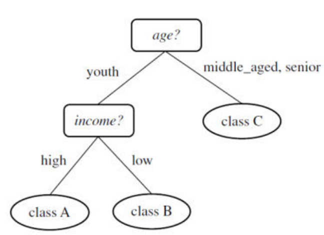
- Predictive analysis is a method used to predict future outcomes based on past data
- The techniques used for prediction are
    - Classification
    - Regression

Classification :

- Classification is used when the outcomes belongs to specific categories (or) groups
- It uses methods like **IF-THEN**, **decision trees**, neural networks to predict a class (or) essentially classify a collection of items

Ex :

$$age(X, \text{"youth"}) \ AND \ income(X, \text{"high"}) \longrightarrow class(X, \text{"A"})$$
$$age(X, \text{"youth"}) \ AND \ income(X, \text{"low"}) \longrightarrow class(X, \text{"B"})$$
$$age(X, \text{"middle\_aged"}) \longrightarrow class(X, \text{"C"})$$
$$age(X, \text{"senior"}) \longrightarrow class(X, \text{"C"})$$



Regression :

- Regression is used when the outcome is a continuous number rather than a category

Ex :

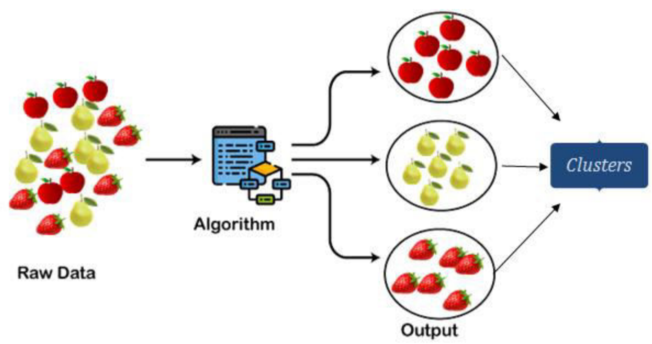
- Suppose a house price prediction model learns from past house sales and predicts the price of a new house based on factors like size, location and number of rooms
- Predicting a student's future exam score based on study hours
- Predicting the temperature for tomorrow based on past weather data

Cluster analysis :

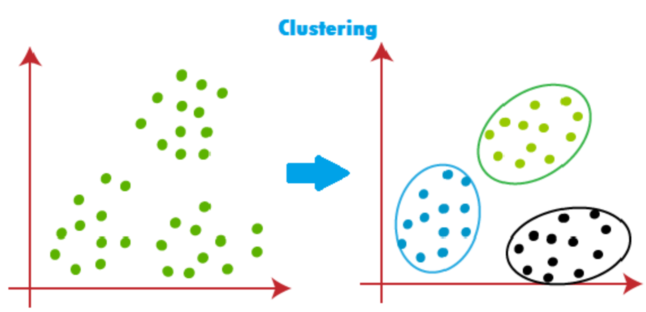
- Cluster analysis in data mining is a method of grouping similar data points together
- It helps find patterns in data by organizing similar items into clusters without using pre-defined labels
- Some common clustering algorithms are K-means, Hierarchical clustering, DBSCAN and so on

Ex :

- Imagine sorting a collection of mixed fruits into groups of apples, strawberries and lemons based on their similarities



- Similarly clustering groups similar data together in a meaningful way

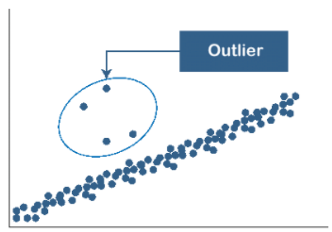


Outlier analysis :

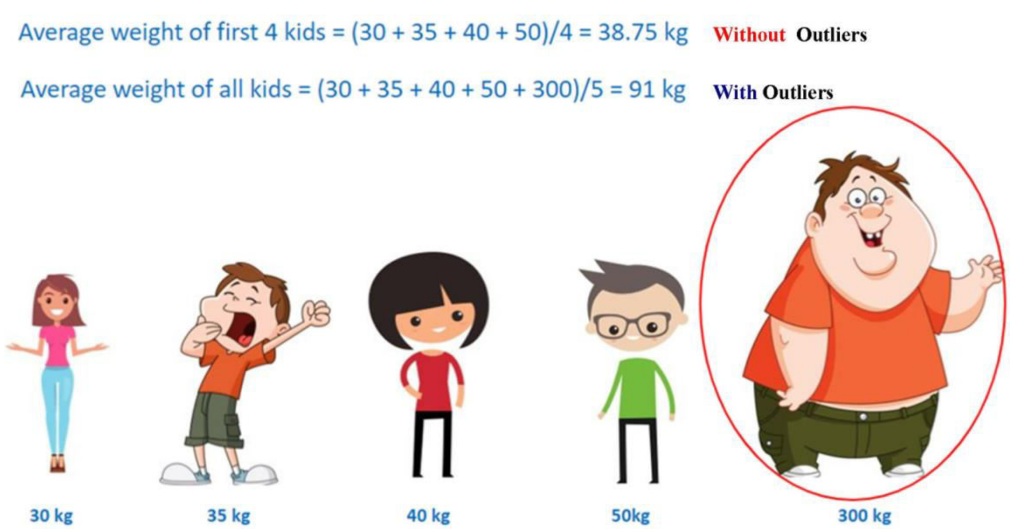
- Outlier analysis in data mining is the process of identifying data points that are very different from the rest of the data
- These unusual data points are called outliers
- Outliers may indicate errors, fraud and rare events

Ex :

- Imagine a classroom where most students are between 10 and 12 years old
- If one student is 18 years old, they are an outlier because their age is very different from the rest



Note :



Integration of data mining system with a database or data ware house system :

- Integrating a data mining system with a database (or) data warehouse means connecting the data mining software directly to where the data is stored
- This makes it easier to access, analyze and extract useful patterns from the data
- In data mining system integration, different schemes define how the data mining system is connected to a database (or) data warehouse
- They are 4 types of schemes
    - No coupling
    - Loose coupling
    - Semi-tight coupling
    - Tight coupling

No coupling :

- The data mining system is independent and it does not use a database
- Data is manually collected and processed separately
- For example, a researcher collects data in excel and runs a separate software tool for analysis

Loose coupling :

- The data mining system and database are separate, but they can exchange data
- The system retrieves data from the database when needed, but processing happens externally
- For example, a company exports sales data from a database and then loads it into a data mining tool like Weka (or) python for analysis

Semi-tight coupling :

- Some data mining operations use the databases processing power [i.e., SQL queries]
- The system can store intermediate results in a database
- For example, a customer analysis tool uses SQL queries to fetch relevant data, then processes it further using advanced mining algorithms

Tight coupling :

- The data mining system is fully integrated into the database (or) data warehouse
- The mining algorithms run directly inside the database
- For example, a retail store uses an integrated system where the database automatically finds shopping trends without exporting data elsewhere

Note :

- No coupling – Data mining system (100%) + Database or Data warehouse (0%) used
- Loose coupling – Data mining system (100%) + Database or Data warehouse (20%) used
- Semi-tight coupling - Data mining system (100%) + Database or Data warehouse (70%) used
- Tight coupling - Data mining system (100%) + Database or Data warehouse (100%) used

- Tight coupling is most efficient and loose coupling and semi-tight coupling are easy to setup if the database and mining tools are separate

Note :

Coupling type	Explanation	Example (Online book store)
No coupling (Standalone system)	Data mining system is separate from the database. Manually data collection and analysis	The bookstore owner keeps sales records in excel and manually checks which books are selling the most
Loose coupling	Data mining system and database are separate but can exchange data. Data is exported from the database for analysis	The manager downloads sales data from the database and imports it into power BI to see trends
Semi-tight coupling	Some processing is done in the database before further analysis in an external system	The manager runs an SQL query to fetch book sales for the last 7 months, ten uses a machine learning tool to predict the future best sellers
Tight coupling	The data mining system is built into the database and runs automatically and no need to export data	The database automatically detects that "People who buy mystery books also buy crime thrillers" and recommends books to customers in real-time

Types of data sets :

Dimensionality :

- Dimensionality refers to the number of features (or) attributes in a dataset

Ex :

- Imagine you are collecting data about students
- If you record only height and weight then it is a 2 dimensions dataset
- If you also add age, grades and attendance then it is a 5 dimensions dataset

Curse of dimensionality :

- As the number of features (or) dimensionality increases, analyzing the data becomes harder
- This problem is called the curse of dimensionality

Ex :

- Suppose you are looking for a lost key in a small room (low dimensionality) and You can search quickly
- Now, imagine searching for the same key in a huge mansion with 100 rooms (high dimensionality) and It becomes much harder to find the key
- Similarly in data analysis, when there are too many attributes, patterns become difficult to detect because the data becomes to spread out

Sparsity : (Sparse data)

- A dataset is called sparse when most of its values are zero (or) empty

Ex :

- Imagine a supermarket tracking what customers buy

Customer	Apples	Bananas	Milk	Bread	Chocolate
John	1	0	0	1	0
Emma	0	0	1	0	1
Alex	0	1	0	0	0

- Each row represents a customer
- Each column represents a product
- 1 means the customer bought that item and a 0 means they didn't
- Most values are 0 because a customer buys only a few items from thousands available

Resolution :

- Resolution refers to how detailed the data is
- If the data is too detailed, it may contain too much noise
- If it is too broad then important patterns may disappear

Ex :

- If you check the atmospheric pressure every hour, you can see detailed changes like storms moving
- If you check it only once a month, you won't see storms because they last only a few days
- Choosing the right resolution is important to see meaningful patterns

Record data :

- Record data is a type of data where each entry (or) record follows the same structure
- Think of it like a table in excel (or) database where every row is a record and every column is an attribute (or) property of that record

Tid	Refund	Marital Status	Taxable Income	Cheat
1	Yes	Single	125 K	No
2	No	Married	100 K	No
3	No	Single	70 K	No
4	Yes	Married	120 K	No
5	No	Divorced	95 K	Yes
6	No	Married	60 K	No
7	Yes	Divorced	220 K	No
8	No	Single	85 K	Yes
9	No	Married	75 K	No
10	No	Single	90 K	Yes

- Record data are three types

- Transaction (or) market basket data
- Data matrix
- Sparse data matrix

<u>Transaction (or) market basket data :</u>

- Each record represents a transactions, where the number of attributes (or) items may vary
- Common in shopping data, banking transactions and so on

TID	ITEMS
1	Bread, Soda, Milk
2	Beer, Bread
3	Beer, Soda, Diaper, Milk
4	Beer, Bread, Diaper, Milk
5	Soda, Diaper, Milk

<u>Data matrix :</u>

- Data matrix is a way to organize and represent data in a table format
- Each row represents a different object (or) item [i.e., Person, Product]
- Each column represents a specific characteristic (attribute) of those objects such as age, height and weight
- Since the attributes are numerical, we can treat each object as point in a multidimensional space and perform mathematical operations on the data

<u>Note :</u>

- Imagine we are collecting data about three students
- We record their age, height and weight

Student	Age	Height (cm)	Weight (kg)
Alice	20	160	55
Bob	22	170	65
Charlie	21	175	70

- This table can written as a data matrix

$$\begin{bmatrix} 20 & 160 & 55 \\ 22 & 170 & 65 \\ 21 & 175 & 70 \end{bmatrix}$$

- Each row represents a student

- Each column represents an attribute like age, height and weight
- Since this is a numeric matrix, we can use mathematical operations like addition, subtraction and multiplication to analyze (or) transform the data
- For example, we can calculate averages, normalize values (or) apply statistical methods
- This is why the data matrix is a standard way to store and process numerical data in statistics and machine learning

Sparse data matrix :

- Sparse data matrix is also called as document data matrix
- It is a special case of a data matrix in which the attributes are of the same type and are symmetric [i.e., Only non-zero values are important]

	team	coach	play	ball	score	game	win	lost	timeout	season
Document 1	3	0	5	0	2	6	0	2	0	2
Document 2	0	7	0	2	1	0	0	3	0	0
Document 3	0	1	0	0	1	2	2	0	3	0

Graph based data :

- Graph based data is a way of organizing information nodes and edges
- It is useful when data involves relationships like social networks, maps and family tress

Ex :

- Imagine a social network where people are connected as friends
- We have 3 people like Alice, Bob and Charlie
    - Alice is friends with bob
    - Bob is friends with Charlie
    - Alice and Charlie are not directly connected
- We can represent this as a graph

Alice — Bob — Charlie

Ordered data :

- Ordered data is data that follows a specific sequence (or) ranking
- The order of the values is meaningful and changing it would affect the meaning of the data
- They are three types of ordered data
    - Sequential data
    - Sequence data
    - Time series data
    - Spatial data

Ex :

- Day of the week
  Monday → Tuesday → …. [i.e., the order matters, Monday always comes before Tuesday]

Sequential data :

- Sequential data is also called as temporal data
- Sequential data is an extension of record data, where each record has a time associated with it
- Consider a retail transaction data set that also stores the time at which the transaction took place

Time	Customer	Items Purchased
t1	C1	A, B
t2	C3	A, C
t2	C1	C, D
t3	C2	A, D
t4	C2	E
t5	C1	A, E

Customer	Time and Items Purchased
C1	(t1: A,B) (t2:C,D) (t5:A,E)
C2	(t3: A, D) (t4: E)
C3	(t2: A, C)

Sequence data :

- Sequence data consists of a data set that is a sequence of individual entities such as a sequence of words (or) letters
- It is quite similar to sequential data, except that there is no time stamps
- Instead there are positions in an ordered sequence
- For example, the genetic information of plants and animals can be represented in the sequences of nucleotides that are known as genes

```
GGTTCCGCCTTCAGCCCCGCGCC
CGCAGGGCCCGCCCCGCGCCGTC
GAGAAGGGCCCGCCTGGCGGGCG
GGGGGAGGCGGGGCCGCCCGAGC
CCAACCGAGTCCGACCAGGTGCC
CCCTCTGCTCGGCCTAGACCTGA
GCTCATTAGGCGGCAGCGGACAG
GCCAAGTAGAACACGCGAAGCGC
TGGGCTGCCTGCTGCGACCAGGG
```

Time-Series Data :

- This is data that is collected over time, often at regular intervals

Ex :

- Stock prices, weather data, or monthly sales numbers.

Spatial Data :

- This data represents information about physical locations and their properties.

Ex :

- GPS coordinates location data from smart phones (or) maps

Types of attribute values :

Attribute :

- An attribute is a characteristic (or) feature of an object that we use to describe it
- For a customer, object attributes can be customer_id, customer_name, customer_address and so on

Types of attributes :

- This is the first step of data pre-processing
- We differentiate between different types of attributes and then pre-process the data
- They are two types of attributes
    - Qualitative attributes
    - Quantitative attributes

Qualitative attributes :

- Qualitative attributes are also called as categorical attributes
- Qualitative attributes describes qualities (or) characteristics of an object
- These attributes do not have numerical values and cannot measured with numbers
- Instead they are labels, names, categories that describe something
- Qualitative attributes are three types
    - Nominal attribute
    - Ordinal attribute
    - Binary attribute

Nominal attribute :

- Nominal attributes are just names (or) labels
- They do not have any order [i.e., One is not greater than the other]

Ex :

- Fruit types – Apple, Banana, Orange [i.e., Fruit types is the label]
- Car brands – Tesla, BMW, Audi [i.e., Car brands is the label]

Note :

- We cannot compare them (or) say one higher or lower than the other

Ordinal attribute :

- These have a meaningful order (or) ranking but the differences between them are not measurable
- We can compare them but we cannot measure exact differences

Ex :

- Education level – High school < Bachelor's < Master's < PhD
- Customer satisfaction – Poor < Fair < Good < Excellent
- Clothing size – Small < Medium < Large < Extra large

Note :

- We can rank them but we don't know exactly how much bigger (or) smaller one is compared to another

Note :

Type	Can be Ordered?	Can be Measured?	Example
Nominal	No	No	Colors, Car Brands, Blood Types
Ordinal	Yes	No	Education Level, Movie Ratings, Clothing Size

Binary attribute :

- A binary attribute is an attribute that has only two possible values
- These values are usually represented as
    - Yes / No
    - True / False
    - 1 / 0
- Binary attributes are two types. They are
    - Symmetric attributes
    - Asymmetric attributes

Symmetric attributes :

- Both values are equally important and have no special meaning over each other

Ex :

- Gender – Male(0) and Female (1) [i.e., Both are just different values, neither is more important]
- On / Off – On(1) and Off(0) [i.e., Both are just states of a switch]

Asymmetric attributes :

- One value is more important than the other
- These are common in rare event detection

Ex :

- Fraud transaction – No(0) and Yes(1) → Yes is more important for fraud detection [i.e., Fraud is rare, but detecting it is crucial]
- Spam Email – No (Not spam) and Yes (Spam) [i.e., Spam emails are fewer but need filtering]

Quantitative attributes :

- Quantitative attribute is an attribute that has numerical values and can be measured (or) counted
- These attributes represent quantities and allow us to perform mathematical operations like addition, subtraction and averaging
- Quantitative attributes are two types
    - Discrete attributes
    - Continuous attributes

Discrete attributes : (Countable numbers)

- These take whole number values [i.e., Cannot have decimals]
- They are obtained by counting something

Ex :

- Number of students in a class – 25, 30, 40
- Number of siblings – 0, 1, 2, 3

Continuous attributes : (Measurable numbers)

- These can take any numerical value including decimals and fractions
- They are obtained by measuring something

Ex :

- Height of a person – 172.3 cm, 180 cm

DATA PRE-PROCESSING

Table of contents :

- Measures of similarity and dissimilarity - basics

Data pre-processing :

- Data pre-processing is the process of cleaning, transforming and organizing raw data into a structure and usable format before analysis
- It involves handling missing values, correcting errors, removing duplicates and converting data into a consistent format to improve the accuracy and efficiency of data mining and machine learning algorithms

Why pre-process the data :

- In the real world, many databases and data warehouses have noisy, missing and inconsistent data due to their huge size
- Low quality data leads to low quality data mining

Noisy :

- Noisy data refers to data that contains errors, inconsistencies and irrelevant information that can negatively affect analysis
- Noisy data can come from various sources such as human mistakes, sensor errors and so on
- For example, a dataset of student scores where values are missing (or) wrongly entered like " -50 " when scores should be between 0 and 100

Missing :

- Missing data means that some information is not recorded in a dataset
- This can happen for different reasons such as mistakes, system errors and so on

Ex :

Name	Age	Occupation	Salary
John	28	Engineer	5000
Lisa	32		6000
Mark		Teacher	4000
Anna	25	Doctor	

Inconsistent :

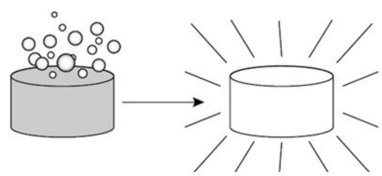
- Data inconsistency meaning is that different versions of the same data appear in different places
- For example, ZIP code is saved in one table as 1234-78 numeric data format, while in another table if may be represented in123478
- Inconsistent data may come from errors in data entry, merging data from different sources with varying formats, differences in the data collection process

Major tasks in data pre-processing :

- Data pre-processing in data mining involves several important steps to clean and organize raw data before analysis
- The major tasks in data pre-processing are
  - Data cleaning
  - Data integration
  - Data reduction
  - Data transformation
  - Data discretization

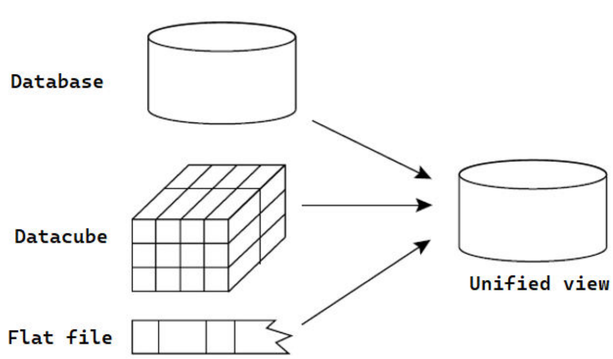
Data cleaning :

- Data cleaning is a process that cleans the data by filling in the missing values, smoothing noisy data, analyzing and removing outliers and removing inconsistencies in the data
- Data cleaning means fixing mistakes, removing wrong (or) missing information and organizing data properly, so it can be used for analysis



Data integration :

- Data integration is the process of combining data from multiple sources into one unified view
- This process involves identifying and accessing the different sources and mapping the data to a common format
- Different data sources may include multiple data cubes, databases, flat files and so on
- The goal of data integration is to make it easier to access and analyze data that is spread across multiple systems (or) platforms, in order to gain a more complete and accurate understanding of the data



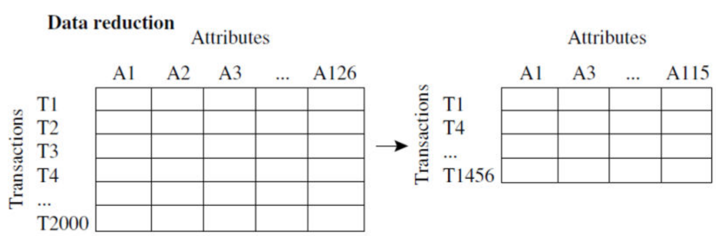
- Data integration strategy is typically described using a triple (G, S, M) approach
- Where G denotes Global schema
  S denotes Schema of the heterogeneous data sources
  M represents the mapping between the queries of the source and global schema

Ex :

- To understand the (G, S, M) approach, let us consider a data integration scenario that aims to combine employee data from 2 different HR databases like database A and database B
- The global schema (G) would define the unified view of employee data including attributes like Employee ID, Name, Department and Salary
- In the schema of heterogeneous sources, database A (S1) might have attributes like Emp ID, Full Name, Dept and Pay
- While database B (S2) might have attributes like ID, Employee Name, Department Name and Wage
- The mappings (M) would then define how the attributes in S1 and S2 map to the attributes in G, allowing for the integration of employee data from both systems into the global schema

Data reduction :

- Data reduction is a technique used in data mining to reduce the size of a dataset while still preserving the most important information
- This can be beneficial in situations where the dataset is too large to be processed efficiently (or) where the dataset contains a large amount of irrelevant (or) redundant information

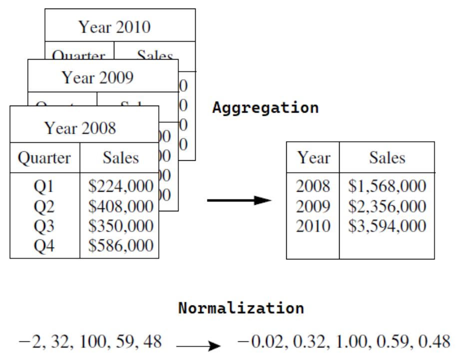


Ex :

- Imagine you have a school database with 10,000 students and 50 attributes like name, age, address, parent's details, hobbies, grades and so on
- But for a report, you only need the student's name, class and average grade
    - Instead of keeping all 50 attributes, you keep only 3 important ones like name, class and average grade [i.e., Dimensionality reduction]
    - Instead of storing data for all 10,000 students, you take a sample of 1,000 students to analyze trends [i.e., Numerosity reduction]
    - Instead of storing full addresses, you only keep ZIP codes to save space [i.e., Data compression]
- Now your dataset is now smaller, faster to analyze and easier to understand

Data transformation :

- Data transformation in data mining refers to the process of converting raw data into format that is suitable for analysis and modelling
- The goal of data transformation is to prepare the data for data mining, so that it can be used to extract useful insights and knowledge



Data discretization :

- Data discretization is the process of converting continuous data (which has many possible values) into smaller, fixed categories to make data easier to analyze

Ex :

- Imagine you have a dataset of people's ages and you want to group them into categories instead of dealing with every possible number
- Before discretization (Continuous data)

Name	Age
Alice	23
Bob	35
Charlie	42
David	51
Emma	67

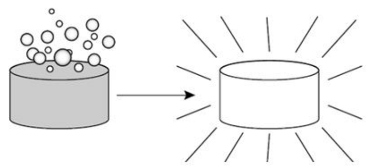
- After discretization (Categorized data)

Name	Age Group
Alice	Young (20-30)
Bob	Middle-aged (31-40)
Charlie	Middle-aged (41-50)
David	Senior (51-60)
Emma	Elderly (61+)

Data cleaning :

- Data cleaning is a process that cleans the data by filling in the missing values, smoothing noisy data, analyzing and removing outliers and removing inconsistencies in the data

- Data cleaning means fixing mistakes, removing wrong (or) missing information and organizing data properly, so it can be used for analysis



- The basic methods of data cleaning are
    - Missing values
    - Noisy data
    - Data cleaning as a process

Missing values :

- When analyzing sales and customer data for All Electronics, You may find that some important details like customer's income are missing
- They are several ways to handle this problem depending on the situation
    - Ignore the tuple
    - Filling in the missing value manually
    - Use a global constant to fill in the missing value
    - Use the attribute mean (or) median to fill in the missing value

Ignore the tuple :

- This is usually done when the class label is missing
- This method is not very effective, unless the tuple contains several attributes with missing values
- For example, if a customer's income, age and location are all missing, You might decide to delete that record entirely

Filling in the missing value manually :

- You can look at other data and decide what the missing value should be
- For example, if you know that most customers in a certain region earn around Rs.50,000, You might enter that value manually

Note :

- It is suggestible for small data sets only
- It is not suggestible for large data sets only because it is time consuming

Use a global constant to fill in the missing value :

- Replace missing values with a fixed label like " Unknown " (or) " -∞ "
- For example, if a customer's income is missing, you can simply write " Unknown "

Use the attribute mean (or) median to fill in the missing value :

- Fill in the missing values with the average (mean) (or) middle (median) value of that attribute
- For example, if the recorded incomes of other customers are Rs.40,000, Rs.50,000 and Rs.60,000, the mean is Rs.50,000 and you can replace the missing income with this value

Noisy data :

- When working with data, sometimes there are random errors (or) fluctuations (noise) that can make it harder to see clear patterns
- Data smoothing helps to reduce this noise, making the data more meaningful
- One common smoothing technique is binning, where we group values into " bins " and adjust them in different ways
- The smoothing can be done in 3 different ways
    - Smoothing by bin means
    - Smoothing by bin medians
    - Smoothing by bin boundaries

Smoothing by bin means :

- Find the mean (average) of each bin and replace all values in the bin with that mean

Smoothing by bin medians :

- Find the median (middle value) of each bin and replace all values with that median

Smoothing by bin boundaries :

- Replace each value with the closest boundary value [i.e., Smallest (or) Largest in the bin]

Ex :

- Imagine you have a set of numbers that represent the prices (in dollars) of a product sold in a store
- The original data(sorted) is

  4, 8, 15, 21, 22, 24, 30, 33, 38
- Now, we divide these numbers into bins (groups)
    - Bin1 : (4, 8, 15)
    - Bin2 : (21, 22, 24)
    - Bin3 : (30, 33, 38)

Smoothing by bin means :

- Bin1 mean : (4 + 8 + 15) / 3 = 9
- Bin2 mean : (21 + 22 + 24) / 3 = 22
- Bin3 mean : (30 + 33 + 38) / 3 = 33.67 (Round to 34)
- New smoothed data is

  9, 9, 9, 22, 22, 22, 34, 34, 34

Smoothing by bin medians :

- Bin1 median : 8 [i.e., Middle of 4, 8, 15]
- Bin2 median : 22 [i.e., Middle of 21, 22, 24]
- Bin3 median : 33 [i.e., Middle of 30, 33, 38]
- New smoothed data is

  8, 8, 8, 22, 22, 22, 33, 33, 33

Smoothing by bin boundaries :

- Bin1 boundaries : 4 (Min), 15(Max)
- Bin2 boundaries : 21 (Min), 24(Max)
- Bin3 boundaries : 30 (Min), 38(Max)
- New smoothed data is

  4, 4, 15, 21, 21, 24, 30, 30, 38

Regression :

- Data smoothing can also be done by regression, a techniques that used to predict the numeric values in a given data set
- It analyses the relationship between a target variable (dependent) and its predictor variable (independent)
- Regression is a form of supervised machine learning technique that tries to predict any continuous values attribute
- Regression can be done in two ways
  - Linear regression
  - Multiple linear regression

Linear regression :

- It uses one variable to predict another [i.e., Ad spending → Sales]

Multiple linear regression :

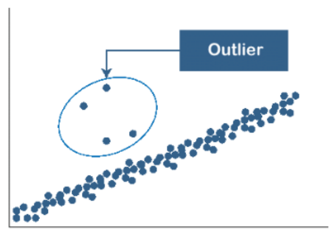
- It uses multiple variables for better predictions [i.e., Ad spending + store size → sales]

Outlier analysis :

- Outlier analysis in data mining is the process of identifying data points that are very different from the rest of the data
- These unusual data points are called outliers
- Outliers may indicate errors, fraud and rare events

Ex :

- Imagine a classroom where most students are between 10 and 12 years old
- If one student is 18 years old, they are an outlier because their age is very different from the rest



Data cleaning as a process :

- Data cleaning is like cleaning up a messy room
- Just like you organize your things, remove unwanted stuff and fix broken items
- Data cleaning helps make data accurate, complete and consistent, so that it can be caused effectively

Finding mistakes (Discrepancy detection) :

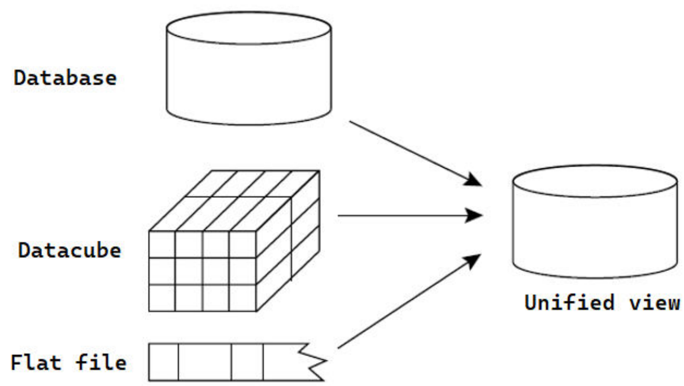
- Mistakes in data can come from many sources such as
    - People making typing errors when entering the data
    - Different formats used for the same information like writing a date as " 2025/02/09 " vs. " 09-02-2025 "
    - Missing values like leaving the " Phone number " filed empty
    - Outdated (or) incorrect information like an old address

Fixing mistakes (Data transformation) :

- Once mistakes are found then we need to fix them
    - Filling in missing values with appropriate data
    - Converting different formats into a standard format like making all dates look the same
    - Removing duplicate (or) incorrect data

Data integration :

- Data integration is the process of combining data from multiple sources into one unified view
- This process involves identifying and accessing the different sources and mapping the data to a common format
- Different data sources may include multiple data cubes, databases, flat files and so on
- The goal of data integration is to make it easier to access and analyze data that is spread across multiple systems (or) platforms, in order to gain a more complete and accurate understanding of the data



Entity identification problem :

- When combining data from different sources likes databases, spreadsheets, files and so on
- We need to make sure that the same real-world things like customers, products, orders are correctly matched
- This is called the entity identification problem because different sources may refer to the same thing using different names, formats and so on

Ex :

- Imagine you are merging customer data from two different companies
- Database 1
    - Customer ID : 12345
    - Name : John Doe
    - Email : Johndoe@gmail.com
- Database 2
    - Customer ID : JD-789
    - Name : Jonathan Doe
    - Email : Johndoe@gmail.com

- These records look different but by examining the email, we can guess they refer to the same person
- Computer program must figure this out automatically - this is the entity identification problem

Note :

- If we don't solve the entity identification problem then we get duplicate records, incorrect reports and making wrong decisions

Redundancy and correlation analysis :

Redundancy :

- Redundancy means storing the same information multiple times unnecessarily
- This can happen when data is repeated (or) when one piece of data can be calculated from other

Ex :

- Imagine you have a customer database with the following information

Customer Name	Birth Date	Age
Alice Smith	15-05-90	33
Bob Johnson	22-09-85	38

- Here, Age is redundant because it can be calculated from the birth date
- Instead of storing both, we can remove Age and calculate it when needed

Correlation analysis :

- Correlation analysis checks if two attributes (columns) are related
- If two columns always give the same information then one of them is unnecessary and can be removed

Ex :

- If a database has " Salary in dollars " and " Salary in rupees ", then one of them is redundant because you can always calculate one form the other
- Correlation analysis will detect that these two attributes are strongly related, so we can keep only one to avoid redundancy

Note :

- Reduce storage space by removing unnecessary data
- Improves data quality by avoiding inconsistencies
- Making data processing faster by reducing the number of columns

Tuple duplication :

- Tuple duplication happens when the same data appears multiple times in a table
- This can happen because of mistakes in data entry and so on

Ex :

- Imagine a customer database for an online store

Customer_ID	Name	Address	Phone
101	John	NY, USA	1234567890
102	Jane Doe	LA, USA	9876543210
103	John	NY, USA	1234567890

- In the above table you have observed that john appears twice with the same address and phone number
- This is a duplicate tuple and should be removed to avoid confusion

Data value conflict detection and resolution :

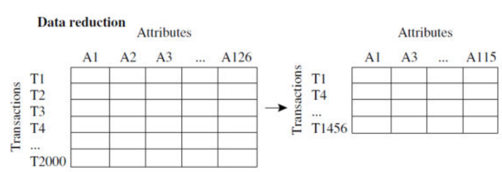
- When combining data from different sources, there can be conflicts and differences in the way information is stored
- These differences need to be detected and resolved to make the data consistent and useful

Ex :

- Different measurement units [i.e., One system records weight in Kilograms (kg), while another uses pounds (lbs). If we don't convert them properly the data won't make sense]
- Different currencies and services
- Different grading systems in schools

Data reduction :

- Data reduction is a technique used in data mining to reduce the size of a dataset while still preserving the most important information
- This can be beneficial in situations where the dataset is too large to be processed efficiently (or) where the dataset contains a large amount of irrelevant (or) redundant information



Overview of data reduction strategies :

- Data reduction strategies are three types. They are
  - Dimensionality reduction
  - Numerosity reduction
  - Data compression

Dimensionality reduction :

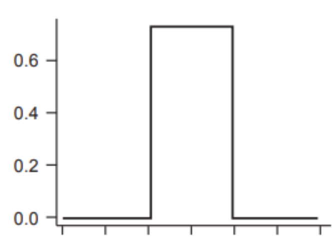
- Dimensionality reduction is a technique in data mining that reduces the number of features (columns (or) attributes) in a dataset while keeping the most important information
- Dimensionality reduction helps remove unnecessary information, making data easier to process, visualize and analyze while preserving important patterns
- It helps make data easier to process, visualize and analyze without losing key patterns
- The dimensionality reduction techniques are
  - Wavelet transforms
  - Principal component analysis
  - Attribute subset selection

Wavelet transforms :

- Discrete wavelet transform is a technique used in signal processing to transform data into a different form, making it easier to analyze, compress data efficiently and remove noise
- Imagine you have a big, messy picture and you want to see only the important details
- Instead of looking at the entire image at once, you zoom in on different parts
- DWT does something similar by breaking down data into smaller, more useful pieces
- Discrete wavelet transform splits data into two ways
  - Approximation information (or) Low frequency
  - Detailed information (or) High frequency

Approximation information :

- This is the simplest wavelet, also called Haar-2
- It looks like a step function (or) a square shape
- It can detect sharp changes in signals as shown in the below figure

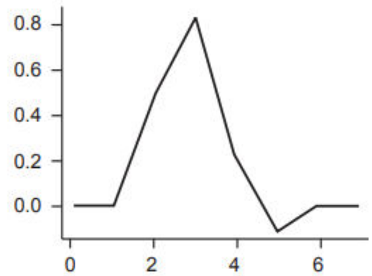


(a) Haar - 2

- For example, if you take an image and use the Haar-2 wavelet, it can highlight edges (or) sudden brightness changes

Detailed information :

- This is more advanced wavelet (Daubechies-4)
- It has a smoother and more complex shape
- It is better for capturing gradual changes in signals as shown in the figure



(b) Daubechies - 4

- For example, if you use it on an audio signal, it can help filter noise while keeping the important details

Note :

Haar-2 wavelet :

- Imagine you are turning a light switch on and off
- If the light is off (0 brightness) and then suddenly on (1 full brightness) that sharp change represents a Haar-2 wavelet

Daubechies-4 wavelet :

- Imagine turning up the volume on your phone gradually
- Instead of an instant jump from 0 to full volume, the volume increases smoothly over time

Note :

Wavelet transforms :

- Discrete wavelet transform is a technique used in signal processing to transform data into a different form, making it easier to analyze, compress and remove noise
- Think of a **data vector X** as a list of numbers, like this

$$X = (5, 10, 15, 20, 25, 30, 35, 40)$$

- When we apply DWT, it transforms this list into **wavelet coefficients** (another list of numbers) that represent the same information in a different way

Ex :

- Let's take a simple example with four numbers

$$X = (8, 6, 5, 3)$$

- Smoothing (or) Averaging
  - Find the average of each pair

$$\frac{8+6}{2} = 7, \quad \frac{5+3}{2} = 4$$

  - This gives a new smoothed version – (7, 4)
- Detail (or) difference calculation
  - Find the difference between each pair

$$8 - 6 = 2, \quad 5 - 3 = 2$$

  - This captures the details (high-frequency changes) - **(2, 2)**
- Now repeat the process for (7, 4)
  - Smoothing (or) Averaging

$$\frac{7+4}{2} = 5.5$$

  - Detail (or) difference calculation

$$7 - 4 = 3$$

  - Now, we get the final transformed vector

$$(5.5, 3, 2, 2)$$

Principal component analysis :

- Principal Component Analysis (PCA) is a technique used in data mining and machine learning to reduce the number of variables (or features) in a dataset while keeping as much important information as possible
- It helps simplify complex data, making it easier to analyze and visualize

Ex :

- Imagine we have data for 3 students, showing their height (in cm) and weight (in kg)

Student	Height (cm)	Weight (kg)
A	160	55
B	170	65
C	180	75

- We have two variables like height and weight
- However, these two are closely related (taller people usually weigh more)
- PCA will combine them into a single new variable (Principal Component) to simplify analysis

- Now, instead of using two variables (Height, Weight), we use one new variable

Student	Principal Component (Body Size)
A	1.2
B	0
C	-1.2

- This new variable (Principal Component 1) still represents the students well, but in a simpler way

Attribute subset selection :

- Attribute subset selection is the process of selecting only the most important attributes (or) features from a dataset and removing unnecessary ones
- This helps to reduce complexity, improve accuracy and speed up data analysis

Ex :

- Imagine you are a teacher trying to predict whether a student will pass (or) fail based on a dataset with these attributes

Student	Age	Height (cm)	Weight (kg)	Study Hours	Attendance (%)	Previous Scores	Pass/Fail
A	16	160	55	3	80%	70	Pass
B	17	170	65	1	50%	40	Fail
C	18	175	75	4	90%	85	Pass

- Some attributes like height and weight may not be relevant to predicting whether a student will pass or fail
- Attribute subset selection helps us keep only the most useful attributes such as study hours, attendance and previous scores
- Remove the irrelevant attributes like age, height and weight

Methods for attribute selection :

- There are four main methods used to select the best attributes (or) features in data mining
    - Stepwise forward selection
    - Stepwise backward elimination
    - Combination of forward selection and backward elimination
    - Decision tree induction

Stepwise forward selection :

- At start empty set of attributes is considered as reduces set
- The best of original attributes are added is this set
- This process is iterated

Initial attribute set :

$\{A_1, A_2, A_3, A_4, A_5, A_6\}$

Initial reduced set :

$\{\,\}$

$\Rightarrow \{A_1\}$

$\Rightarrow \{A_1, A_4\}$

$\Rightarrow$ Reduced attribute set :

$\{A_1, A_4, A_6\}$

Stepwise backward elimination :

- With availability of full set of attributes at each step, worst attributes are removed

Initial attribute set :

$\{A_1, A_2, A_3, A_4, A_5, A_6\}$

$\Rightarrow \{A_1, A_3, A_4, A_5, A_6\}$

$\Rightarrow \{A_1, A_4, A_5, A_6\}$

$\Rightarrow$ Reduced attribute set :

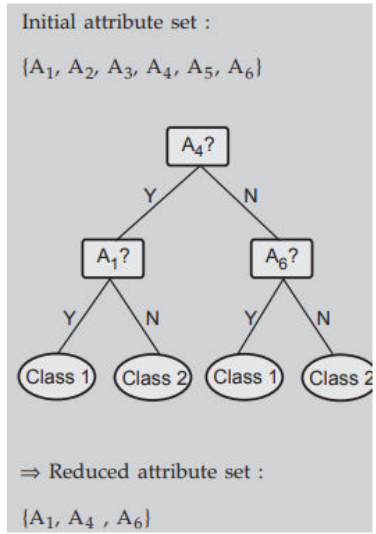
$\{A_1, A_4, A_6\}$

Combination of forward selection and backward elimination :

- In this at each step best attribute is selected as per stepwise forward selection and worst is removed as per stepwise backward elimination
- Thus it combines both the methods

Decision tree induction :

- Flowchart like structure is created where each internal non-leaf node denotes test on an attribute and each branch denotes outcome of the test



Numerosity reduction :

- Numerosity reduction reduces the original data volume and represents it in a much smaller form
- The Numerosity reduction techniques are
    - Parametric Numerosity reduction
    - Non-parametric Numerosity reduction

Parametric Numerosity reduction :

- Parametric Numerosity reduction is a technique used to reduce the size of a dataset by replacing the actual data with mathematical model (Parameters)
- Instead of storing all the data points, we store only a few parameters that describe the data

Note :

- Imagine you have a dataset of 1000 points representing temperatures recorded every hour for a month
- Instead of storing all 1000 values, you can fit a mathematical model like a linear regression equation

  Temperature = 2 x (Hour) + 20
- Now, instead of storing 1000 numbers, you only store two parameters (2 and 20) of the equation
- This reduces the amount of data but still gives a good approximation of the temperature trend

Non-parametric Numerosity reduction :

- Non-parametric Numerosity reduction reduces data without assuming a specific mathematical model
- Instead of replacing data with a formula, it uses different types of techniques are
    - Clustering

- Sampling
- Histograms
- Data cube aggregation

Clustering :

- Clustering is a method used to reduce the amount of data grouping similar data points together and representing them with a few key values instead of storing every individual data point
- This helps in simplifying large datasets while keeping important patterns

Ex :

- Imagine you have a list of 100 students with their heights recorded
- Instead of storing all 100 heights, you group them into 3 clusters
    - Short (Below 5.4 feet)
    - Medium (5.4 to 6 feet)
    - Tall (above 6 feet)
- Now instead of keeping all 100 heights, you can just store
    - The average height for each group
    - The number of students in each group
- This reduces the amount of data while still maintaining useful information about height distribution

Sampling :

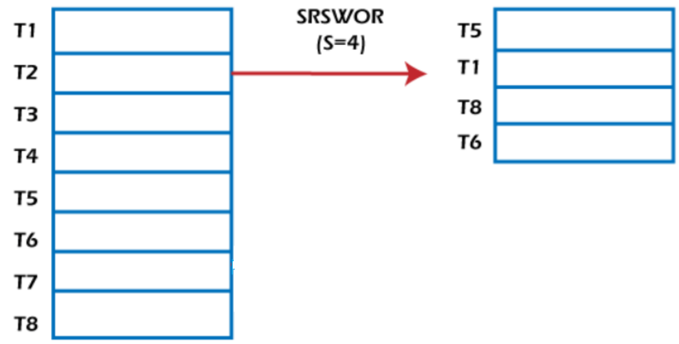
- Sampling in data mining is the process of selecting a small portion (sample) of data from a large dataset
- This helps in analyzing data efficiently without processing the entire dataset, which can be time-consuming and resource-intensive
- Sampling is useful when dealing with big data, where analyzing the full dataset is impractical
- Sampling are classified into three types
    - Simple random sample without replacement (SRSWOR)
    - Simple random sample with replacement (SRSWR)
    - Cluster sample
    - Stratified sample

Simple random sample without replacement (SRSWOR) :

- In simple random sampling without replacement, each item is chosen randomly and once selected, it is not placed back
- This means no item can be selected more than once

Ex :

- Imagine you have a box with 8 slips of paper, each labeled T1, T2, T3, …., T8 as shown in the below figure



- You want to pick slips randomly, but once you pick a slip, you do not pick it back in the box
    - Pick a slip at random – Suppose you pick T5 [i.e., Set it aside]
    - Now you have 7 slips left in the box
    - Pick another slip at random – Suppose you pick T1 [i.e., Set it aside again]
    - Now, only 6 slips remains
    - Repeat this until you have 4 slips – Suppose the next picks are T8 and T6
- Your sample is { T5, T1, T8, T6 }
- Since you did not put back any slips, each selection reduces the number of available options
- This is called a simple random sampling without replacement

<u>Note :</u>

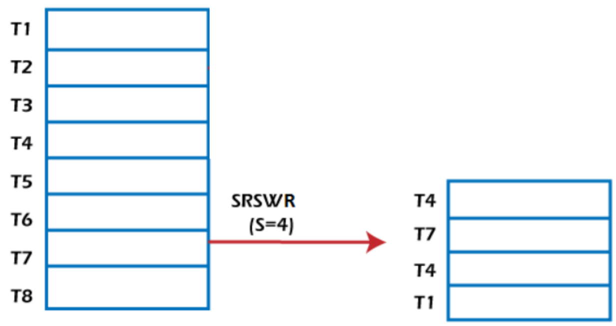
- SRSWOR – Not allowing duplicates
- SRSWR – Allowing duplicates

<u>Simple random sample with replacement (SRSWR) :</u>

- In simple random sampling with replacement, each item is chosen randomly and after selection, it is placed back into the selection tool
- This means the same item can be picked multiple times

<u>Ex :</u>

- Imagine you have a box with 8 slips of paper, each labeled T1, T2, T3, …., T8 as shown in the below figure



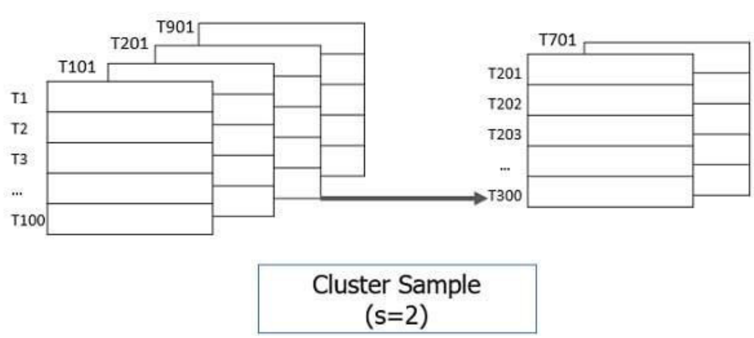
- You want to pick slips randomly, but this time you put the slip back in the box after each pick
  - Pick a slip at random – Suppose you pick T4 → Put it back in the box
  - All 8 slips remain in the box, Since we put T4 back
  - Pick another slip at random – Suppose you pick T7 → Put it back in the box
  - All 8 slips are still available
  - Pick another slip at random – Suppose you pick T4 again [i.e., This is possible because T4 was placed back]
  - Put it back in the box again
  - Pick another slip at random – Suppose you pick T1 → Put it back in the box
- Your sample is { T4, T7, T4, T1 }

Cluster sample :

- Cluster sampling in data mining is a technique where a large dataset is divided into smaller groups (or) clusters
- Only some of these clusters are selected for analysis instead of studying the whole dataset
- This helps in reducing effort and cost while still getting meaningful insights

Ex :

- A large dataset is divided into smaller groups (or) clusters and only a few clusters are selected for analysis as shown in the below figure



- The dataset is divided into multiple clusters like T1,T101,T201, … T901 and so on each containing multiple data points
- There are any clusters and analyzing all of them would be time consuming

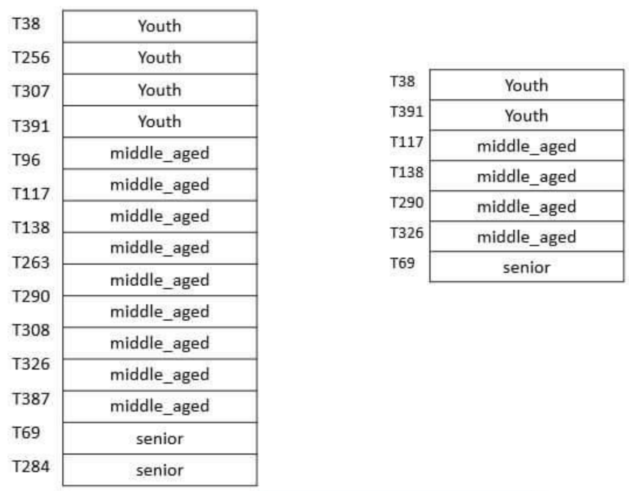
- Instead of selecting individual data points randomly, entire clusters are chosen
- In this case S = 2, meaning two clusters are selected for analysis
- Only some clusters T201, T701 are chosen
- The data from these selected clusters are analyzed, while the rest are ignored

Stratified sample :

- Stratified sampling is a technique where a dataset is divided into different groups (or) strata based on a specific characteristic and then a proportional number of samples are taken from each group
- This ensures that each group is well-represented in the final dataset

Ex :

- Imagine a dataset consists of individuals categorized in to three age groups as shown in the below figure

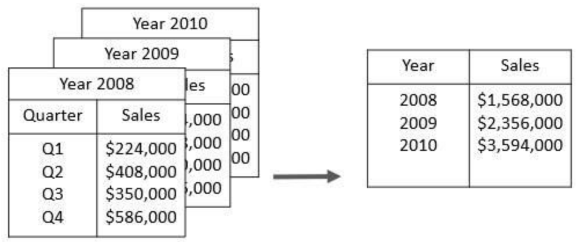


Stratified sample (according to age)

- In the above figure you have observed that three age groups are youth (T38, T256, T307, T391), Middle-aged (T96, T117, T138, T263 and so on), Senior (T69, T284)
- There are more middle-ages individuals compared to other groups
- Instead of randomly selecting individuals, we first divide them into age groups(strata) and then we select a proportional number of samples from each group
- Youth – T38, T391 are the subset from the youth group
- Middle-aged – T117, T138, T290, T326 are subset from the middle-aged group
- Senior - T69 are the subset from the senior group
- Each age group is represented in the final sample

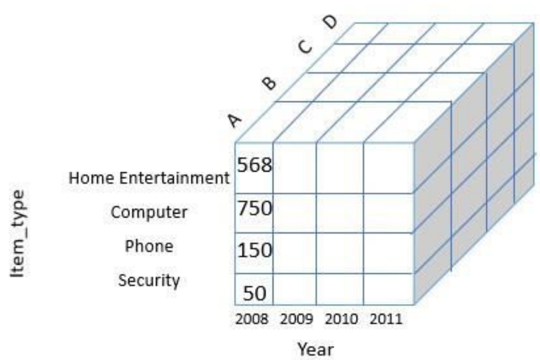
Data cube aggregation :

- This technique is used to aggregate data in simpler form
- for example, suppose you have the data of All Electronics sales per quarter for the year 2008 to the year 2010
- in case you want to get the annual sale per year then you just have to aggregate the sales per quarter for each year
- in this way, aggregation provides you with the required data which is much smaller in size and thereby we achieve data reduction even without losing any data



Aggregated Data

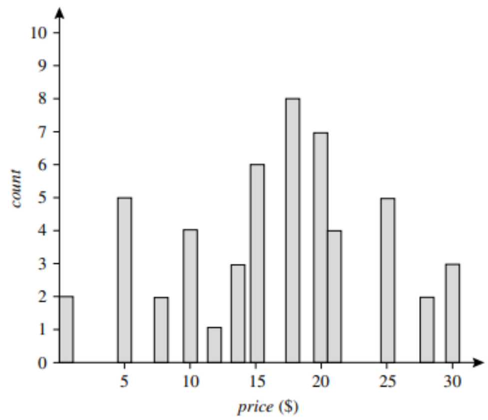
- The data cube aggregation is a multidimensional aggregation which eases multidimensional analysis
- Like in the image below the data cube represent annual sale for each item for each branch
- The data cube present pre-computed and summarized data which eases the data mining into fast access



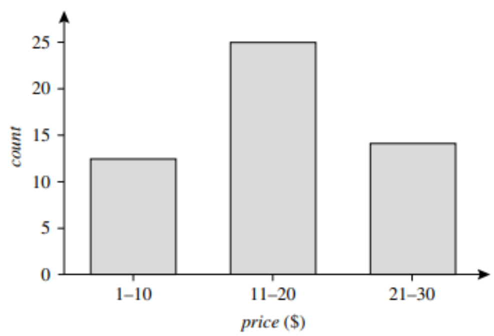
Data Cube Aggregation

Histograms :

- A histogram is a type of graph used in data mining and statistics to show the distribution of a dataset
- It divides the data into bins (or) intervals)0 and shows how many data points fall into each bin
- It helps us understand the frequency (or) how often certain values appear in the dataset



A histogram for *price* using singleton buckets—each bucket represents one price–value/ frequency pair



An equal-width histogram for *price*, where values are aggregated so that each bucket has a uniform width of $10

Data compression :

- Data compression is a technique that reduces the size of data, so that it takes up less space on a computer (or) storage device
- It does this by removing unnecessary parts (or) replacing repetitive parts with shorter codes
- The data compression techniques are
    - Lossless compression (No data loss)
    - Lossy compression (Some data is lost)

Lossless compression :

- Reduces the file size without losing any information
- It is mainly used in ZIP files, PNG images, text compression and so on

Ex :

- Imagine a text " AAAAABBBBCCCC"
- Instead of storing all letters, Run-length encoding (RLE) stores it as " 5A4B4C" meaning " A appears 5 times, B appears 4 times and C appears 4 times "
- When needed, we can reconstruct the exact original text from this compressed form

Lossy compression :

- Reduces file size by removing details that are not very important
- The original data cannot be restored exactly, but it is still useful
- It is mainly used in JPEG images, MP3 music, YouTube videos and so on

Ex :

- Suppose we have to take an example as " JPEG image compression "
    - High quality image might have millions of colors
    - Lossy compression removes some similar colors to make the file smaller
    - The image still looks good, but it is not exactly the same as the original

Data transformation and data discretization :

Data transformation :

- Data transformation in data mining refers to the process of converting raw data into format that is suitable for analysis and modelling
- The goal of data transformation is to prepare the data for data mining, so that it can be used to extract useful insights and knowledge .
- The main steps for data transformation are
    - Smoothing
    - Aggregation
    - Discretization
    - Attribute construction
    - Generalization
    - Normalization
    - Concept hierarchy generation for nominal data

Smoothing :

- Removes random variations in data to make patterns clearer

(or)

- Removing noise and making data clearer

Ex :

- Imagine daily temperature readings like 30, 32, 35, 50, 34, 31
- The value 50 seems like an unusual jump (noise)
- Smoothing techniques like averaging, replace it with a more reasonable value
- New values might look like 30, 32, 34, 33, 34, 31 → A clear trend

Aggregation :

- Combines detailed data into summary values to make analysis easier

Ex :

- A store records daily sales
    - Monday – Rs.1000
    - Tuesday – Rs.1200
    - Wednesday – Rs.900
- Instead of keeping every day's data, aggregation gives a weekly total
    - Total sales for the week – Rs.3100

Discretization :

- Groups continuous values into categories

Ex :

- Instead of storing exact ages (22, 30, 45, 60, 75), we create categories
    - Young (0-30)
    - Middle aged (31-50)
    - Senior (51+)
- This makes it easier to analyze trends [i.e., Seniors buy more medicine]

Attribute construction :

- Creates a new useful attributes from existing ones

Ex :

- Instead of storing height and weight separately, we create BMI (Body Mass Index)

$$BMI = Weight\ (Kg) / Height\ (m^2)$$

- This helps in health predictions like detecting obesity trends

Generalization :

- Generalization is the process of grouping detailed data into broader categories to make it easier to analyze and understand
- It helps simplify complex data by converting low-level (specific) details into high-level(more general) concepts

Ex :

- Suppose we have a dataset with people's ages

Name	Age
John	22
Sarah	25
Mike	40
Emma	65

- If we generalize age into broad categories

Name	Age	Generalized Age
John	22	Young
Sarah	25	Young
Mike	40	Middle-aged
Emma	65	Old

- Now, instead of dealing with exact numbers, we have broad categories like Young, Middle-aged, and Old making the data easier to interpret

Normalization :

- Data normalization is a technique used in data mining to transform the values of a dataset into a common scale
- This is important because many machine learning algorithms are sensitive to the scale of the input features and can produce better results when the data in normalized
- Normalization is used to scale the data of an attribute, so that it falls in a smaller range such as -1.0 to 1.0 (or) 0.0 to 1.0
- It is generally useful for classification algorithms

Need of normalization :

- Normalization is generally required when we are dealing with attributes on a different scale, otherwise it may lead to a dilution in effectiveness of an equally important attribute (on lower scale) because of other attribute having values on larger scale
- In simple words, when multiple attributes are there but attributes have values on different scales, this may lead to poor data models while performing data mining operations
- So they are normalized to bring all the attributes on the same scale

person_name	Salary	Year_of_experience	Expected Position Level
Aman	100000	10	2
Abhinav	78000	7	4
Ashutosh	32000	5	8
Dishi	55000	6	7
Abhishek	92000	8	3
Avantika	120000	15	1
Ayushi	65750	7	5

The attributes salary and year_of_experience are on different scale and hence attribute salary can take high priority over attribute year_of_experience in the model.

Methods of data normalization :

- Data normalization can be done by using three data normalization methods
    - Min-max normalization
    - Z-Score normalization
    - Decimal scaling

Min-max normalization :

- Min-max normalization is a technique that scales data to a fixed range, typically 0 to 1 (or) sometimes -1 to 1
- It transforms each value in the dataset based on the minimum and maximum values of that feature
- The formula of min-max normalization is

$$X_{\text{normalized}} = \frac{X - X_{\min}}{X_{\max} - X_{\min}}$$

- Where X = Original value
  X_{min} = Minimum value of the feature
  X_{max} = Maximum value of the feature

Ex :

- Let's say we have the following dataset representing student test scores out of 100

Student	Score
A	20
B	50
C	75
D	100

- Min score (X_{min}) = 20
  Max score (X_{max}) = 100
- Apply min-max normalization formula for each student

- Student A

$$\frac{20-20}{100-20} = \frac{0}{80} = 0.00$$

- Student B

$$\frac{50-20}{100-20} = \frac{30}{80} = 0.375$$

- Student C

$$\frac{75-20}{100-20} = \frac{55}{80} = 0.6875$$

- Student D

$$\frac{100-20}{100-20} = \frac{80}{80} = 1.00$$

- Normalized scores tables as shown in the below

Student	Original Score	Normalized Score
A	20	0
B	50	0.375
C	75	0.6875
D	100	1

- Now, all the scores are scaled between 0 and 1, making it easier for machine learning models to process them fairly

Z-Score normalization :

- Z-Score normalization is also called as standardization
- It transforms data, so that it has a mean of 0 and a standard deviation of 1
- This makes it easier for machine learning models to handle features with different scales
- The formula for Z-Score normalization is

$$X_{\text{normalized}} = \frac{X-\mu}{\sigma}$$

- Where X = Original value
  μ = Mean (or) Average of the feature
  σ = Standard deviation of the feature

Ex :

- Let's say we have the following dataset representing the students test scores

Student	Score
A	50
B	60
C	70
D	80
E	90

- Calculate mean value

$$\mu = \frac{50 + 60 + 70 + 80 + 90}{5} = \frac{350}{5} = 70$$

- Calculate standard deviation
  - First, find the squared differences from the mean

$$(50 - 70)^2 = (-20)^2 = 400$$
$$(60 - 70)^2 = (-10)^2 = 100$$
$$(70 - 70)^2 = (0)^2 = 0$$
$$(80 - 70)^2 = (10)^2 = 100$$
$$(90 - 70)^2 = (20)^2 = 400$$

  - Now, computer the variance

$$\sigma^2 = \frac{400 + 100 + 0 + 100 + 400}{5} = \frac{1000}{5} = 200$$

  - Finally, take the square root to get the standard deviation

$$\sigma = \sqrt{200} \approx 14.14$$

- Apply Z-Score formula for each student
  - Student A

$$Z = \frac{50 - 70}{14.14} = \frac{-20}{14.14} \approx -1.41$$

  - Student B

$$Z = \frac{60 - 70}{14.14} = \frac{-10}{14.14} \approx -0.71$$

  - Student C

$$Z = \frac{70 - 70}{14.14} = \frac{0}{14.14} = 0$$

  - Student D

$$Z = \frac{80 - 70}{14.14} = \frac{10}{14.14} \approx 0.71$$

- Student E

$$Z = \frac{90 - 70}{14.14} = \frac{20}{14.14} \approx 1.41$$

- Normalized scores tables is shown in the below

Student	Original Score	Z-Score Normalized Score
A	50	-1.41
B	60	-0.71
C	70	0
D	80	0.71
E	90	1.41

<u>Decimal scaling :</u>

- Decimal scaling is a technique that transforms data by dividing each value by a power of 10, so that the absolute value of all data points is less than 1
- The formula for decimal scaling is

$$X_{\text{normalized}} = \frac{X}{10^j}$$

- Where X = Original value

  j = The smallest integer such that all normalized values fall between -1 and 1

  10^j is chosen based on the largest value in the dataset

<u>Ex :</u>

- Suppose we have the following dataset of population numbers (in thousands)

City	Population (in thousands)
A	345
B	678
C	123
D	890

- Find the maximum absolute value
  - The highest population is 890
- Determine j
  - The largest number 890 has 3 digits
  - So, we set j = 3, meaning we divide all values by 10^3 = 1000
- Apply the decimal for each student
  - City A

$$\frac{345}{10^3} = \frac{345}{1000} = 0.345$$

  - City B

$$\frac{678}{10^3} = \frac{678}{1000} = 0.678$$

- City C

$$\frac{123}{10^3} = \frac{123}{1000} = 0.123$$

- City D

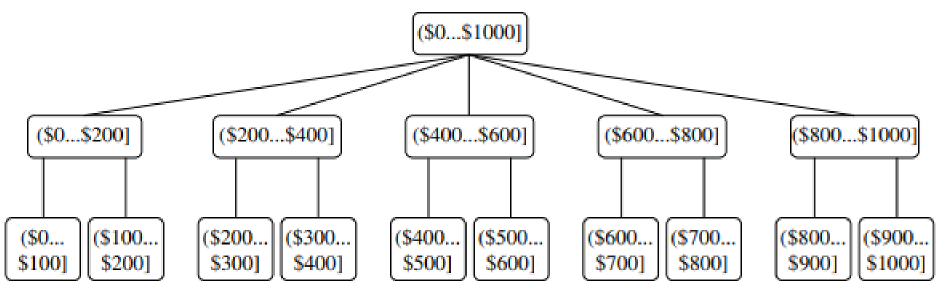
$$\frac{890}{10^3} = \frac{890}{1000} = 0.890$$

- Normalized values table is shown in the below

City	Original Population (in 1000s)	Normalized Population
A	345	0.345
B	678	0.678
C	123	0.123
D	890	0.89

Concept hierarchy generation for nominal data :

- Concept hierarchy organizes data into different levels of abstraction**,** making it easier to analyze by transforming detailed data into summarized data



Data discretization :

- Data discretization is the process of converting continuous data (which has many possible values) into smaller, fixed categories to make data easier to analyze

Ex :

- Imagine you have a dataset of people's ages and you want to group them into categories instead of dealing with every possible number
- Before discretization (Continuous data)

Name	Age
Alice	23
Bob	35
Charlie	42
David	51
Emma	67

- After discretization (Categorized data)

Name	Age Group
Alice	Young (20-30)
Bob	Middle-aged (31-40)
Charlie	Middle-aged (41-50)
David	Senior (51-60)
Emma	Elderly (61+)

Measures of similarity and dissimilarity basics :

Similarity :

- Similarity is a number that tells us how much two things are alike
- High similarity means the objects are more similar
- Lower similarity means they are less similar
- It usually between 0 and 1
- Where 0 means no similarity at all [i.e., Completely different]
  1 means they are exactly the same

Ex :

- Two red apples have a similarity close to 1
- A red apple and a green apple have some similarity but less than 1
- A red apple and a car have a similarity of 0 [i.e., not similar at all]

Dissimilarity :

- Dissimilarity (or) distance is a number that tells us how different two things are
- Lower dissimilarity means the objects are more similar
- Higher dissimilarity means they are more different
- It is usually between 0 and infinity (∞) but sometimes between 0 and 1
- Where 0 means no difference [i.e., Exactly the same]

  A larger number means they are more different

Ex :

- Two identical apples have a dissimilarity of 0
- A red apple and a green apple have some dissimilarity
- A red apple and a car have a very high dissimilarity

Basics :

- Sometimes we need to convert similarity into dissimilarity (or) vice versa
- We may need to adjust their values to fit within a specific range like [0, 1]
- This is important because some software (or) algorithms only work with a certain type of value

Ex :

- If we have similarity values between 1 and 10, but an algorithm only works with values between 0 and 1 then we need to scale the values down
- If an algorithm only works with dissimilarity values then we need to convert similarities into dissimilarities

Convert values to a specific range :

- If we have a similarity that ranges from 1 to 10 then we can transform it to fall within [0, 1] using this formula

$$s' = \frac{s - \min(s)}{\max(s) - \min(s)}$$

- This makes the smallest value 0 and the largest value 1

Ex :

- Suppose our similarity values are 1, 4, 7, 10
- Here min(s) = 1 and max(s) = 10
- Using the formula, the transformed values

$$(1-1)/(10-1) = 0$$
$$(4-1)/(10-1) = 0.33$$
$$(7-1)/(10-1) = 0.67$$
$$(10-1)/(10-1) = 1$$

- Now, the values are in the range [0, 1], which is easier for certain algorithms to process
- Similarly for dissimilarity values, we can use this formula

$$d' = \frac{d - \min(d)}{\max(d) - \min(d)}$$

Transforming dissimilarity from 0 to ∞ :

- If a dissimilarity measure has values between 0 and ∞ then we need a special transformation because we cannot simply divide by a maximum value (since ∞ has no limit)
- One common formula is

$$d' = \frac{d}{1+d}$$

Ex :

- If the original dissimilarities are 0, 0.5, 2, 10, 100, 1000 then

$$0/(1+0) = 0$$
$$0.5/(1+0.5) = 0.33$$
$$2/(1+2) = 0.67$$
$$10/(1+10) = 0.90$$
$$100/(1+100) = 0.99$$
$$1000/(1+1000) = 0.999$$

- As you can see, large values get compressed near 1, making them easier to compare

Converting similarity to dissimilarity :

- If similarity is already in the range [0, 1] then we can convert it into dissimilarity using a simple formula

$$d = 1 - s$$

- And vice versa

$$s = 1 - d$$

Ex :

- If s = 0.8 then d = 1-0.8
  = 0.2
- If d = 0.3 then s = 1-0.3
  = 0.7

- If dissimilarity values go beyond 1 then we need another approach like

$$s = \frac{1}{d+1}$$

- This ensures similarity values stay within [0, 1]

Ex :

If **d = 0**, then $s = \frac{1}{0+1} = 1$ (completely similar)

If **d = 1**, then $s = \frac{1}{1+1} = 0.5$

If **d = 10**, then $s = \frac{1}{10+1} = 0.09$